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Report

On the topic:

Depth-Guided NeRF: Depth-Supervision for Geometrically Consistent 3D  
Scene Reconstruction

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## ABSTRACT

This project investigates the impact of depth supervision on Neural Radiance Field (NeRF) training for geometrically consistent 3D scene reconstruction. We systematically evaluate four supervision strategies: a baseline NeRF trained solely with RGB photometric loss (NONE), a soft depth-supervised NeRF (SOFT) leveraging probabilistic depth guidance, a hard-constrained NeRF (HARD) enforcing explicit free-space and surface-concentration constraints, and a hybrid approach (HYBRID) combining soft regularization with hard geometric constraints. Depth-guided ray sampling, L1 depth consistency loss, and physically motivated volumetric constraints are integrated to improve optimization stability and geometric fidelity. Experiments on the Lego synthetic scene demonstrate that while the baseline achieves the highest peak training PSNR (27.55 dB), it underperforms on unseen views (validation PSNR 19.91 dB) due to overfitting and depth inconsistency. The soft strategy enhances geometric regularization, achieving smoother convergence, improved depth accuracy, and higher validation PSNR (20.72 dB). The hard strategy further enforces geometric constraints, improving depth fidelity and validation PSNR (21.27 dB) at the cost of increased training loss. The hybrid strategy delivers the best overall performance, achieving a peak validation PSNR of (22.32) dB, reduced variance across iterations, sharper surface boundaries, and minimal floating artifacts, effectively balancing geometric precision and visual realism. These findings highlight the critical role of structured depth supervision, particularly hybrid strategies, in improving convergence stability, generalization, and novel-view reconstruction quality for NeRF-based 3D scene modeling, with direct implications for robotics and autonomous perception systems.

# 1 INTRODUCTION

## 1.1 Background and Motivation

The challenge of reconstructing three-dimensional scenes from two-dimensional images has captivated researchers in computer vision and graphics for decades. Traditional approaches relied on explicit geometric representations such as meshes, point clouds, or voxel grids, each carrying inherent limitations in either expressiveness, memory efficiency, or ease of optimization. The introduction of Neural Radiance Fields in 2020 marked a paradigm shift by representing scenes as continuous functions encoded in neural network weights, achieving photorealistic novel view synthesis with unprecedented quality.

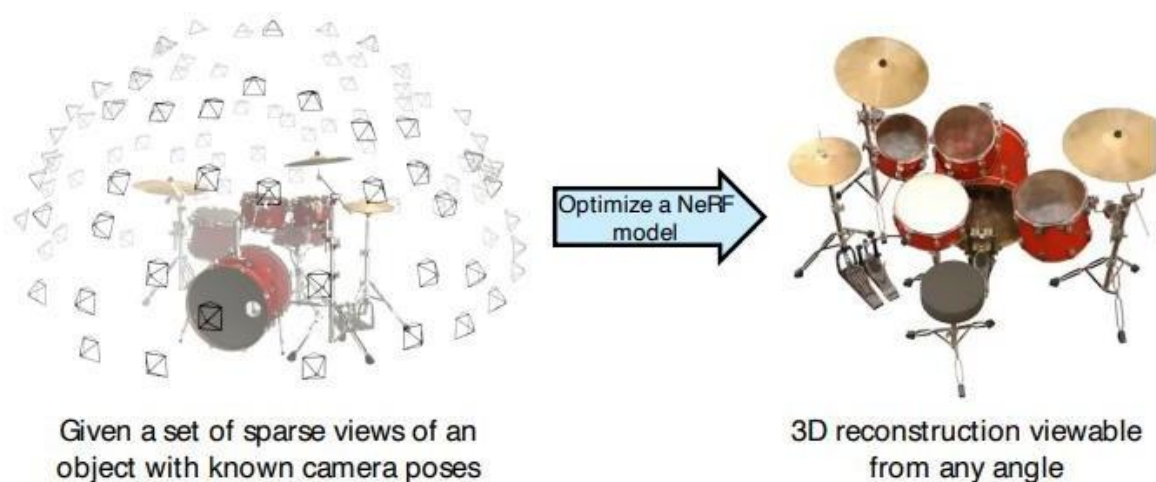


Figure 1 2d image to 3d image reconstruction

Despite the remarkable success of NeRF, the original formulation depends exclusively on photometric consistency across multiple views to learn scene geometry. While this image-only supervision eventually converges with accurate representations, it presents several challenges in practice. The landscape optimization can be complex, with the network required to simultaneously learn both geometric structure and view-dependent appearance from color information alone. This can lead to slow convergence, ambiguity in regions with limited texture or view coverage, and potential failure modes in geometrically challenging scenarios.

Meanwhile, depth sensing technology has become increasingly accessible through various modalities including LiDAR sensors, structured light systems, time-of-flight cameras, and multi-

view stereo algorithms. These sensors provide direct geometric measurements that could potentially guide neural scene reconstruction. The fundamental question motivating this research is whether we can leverage depth information as an additional supervisory signal to improve the training quality, convergence speed, and geometric accuracy of Neural Radiance Fields.

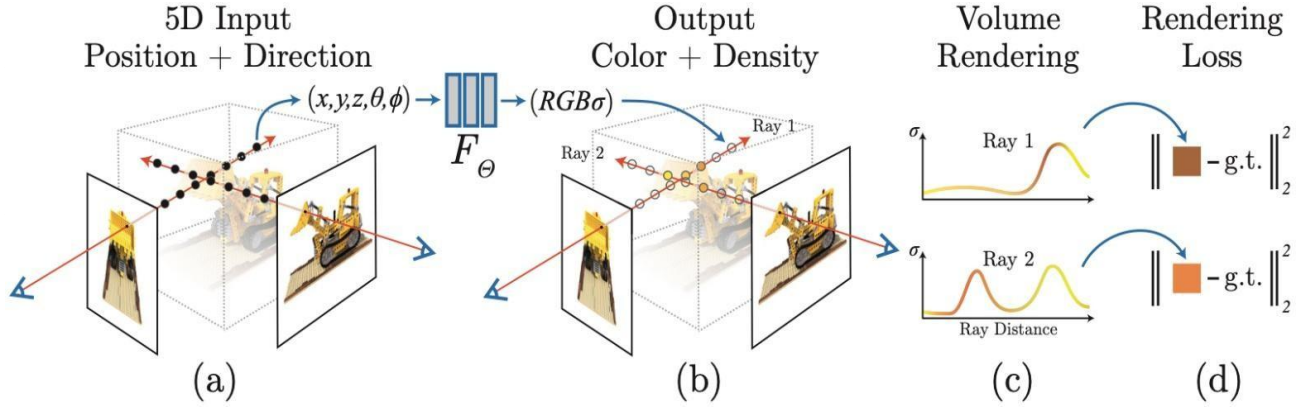


Figure 2 NeRF = volume rendering + coordinate-based network



## 2 Problem Statement

Neural Radiance Fields (NeRF) have emerged as a powerful framework for learning continuous volumetric scene representations from multi-view RGB images, enabling high-quality novel view synthesis. Despite their visual success, the original NeRF formulation relies exclusively on photometric supervision, where the optimization objective minimizes the discrepancy between rendered and ground-truth RGB values. While this design reduces data dependency and simplifies acquisition, it provides only weak and indirect geometric guidance, leading to fundamental limitations in 3D reconstruction quality.

### 2.1 Limitations of Purely Photometric NeRF Training

In photometric-only NeRF training, depth and geometry are not explicitly supervised but instead inferred implicitly through multi-view colour consistency. This inverse problem is inherently ill-posed: multiple depth configurations can produce similar photometric observations, particularly in texture less regions, repetitive patterns, reflective surfaces, or narrow camera baselines. Therefore, the learned volumetric density field may converge to visually plausible yet geometrically inconsistent solutions.

Moreover, photometric supervision entangles appearance and geometry during optimization. The network must simultaneously learn radiance and density without explicit structural constraints, resulting in unstable learning dynamics that are highly sensitive to initialization, sampling strategy, and optimization hyperparameters. These issues manifest as floating density artifacts, blurred or thick surfaces, and delayed stabilization of geometry.

### 2.2 Depth Ambiguity and Slow Convergence in Baseline NeRF

A direct outcome of missing geometric supervision is depth ambiguity. Since baseline NeRF models do not explicitly constrain depth, the density field often evolves slowly and inconsistently along camera rays, especially during early and mid-training phases. Although acceptable rendering quality may be achieved relatively early, the underlying geometry frequently remains inaccurate or under-constrained.

This behaviour leads to slow convergence, requiring many training iterations to achieve stable and physically meaningful geometry. Such inefficiency increases computational cost and limits scalability. More critically, it renders baseline NeRF unsuitable for downstream tasks such as robotic inspection, navigation, and measurement where accurate depth and surface localization are mandatory rather than optional.

## 2.3 Motivation for Depth-Supervised NeRF

Modern perception systems, particularly in robotics, typically integrate RGB cameras with depth sensors such as LiDAR, stereo cameras, or structured-light devices. These sensors provide direct geometric measurements that can serve as valuable supervisory signals. The core challenge, however, lies not in the availability of depth data, but in how depth supervision is integrated into NeRF training.

Different depth supervised NeRF strategies impose varying degrees of geometric constraint, each introducing distinct trade-offs between flexibility, robustness, stability, and geometric fidelity. This work identifies and studies four representative training paradigms:

- Baseline NeRF (None): Purely photometric supervision with no depth guidance.
- Soft Depth-Supervised NeRF: Depth incorporated as a softly weighted regularization term.
- Hard Depth-Constrained NeRF: Depth enforced via strict geometric rules governing free space and surface occupancy.
- Hybrid Depth-Supervised NeRF: A combined strategy that adaptively balances soft regularization and hard constraints.

## 2.4 Soft Depth Supervision: Benefits and Limitations

Soft depth supervision augments the photometric loss with a depth consistency term that penalizes deviations between predicted and measured depth values. This approach improves convergence speed and stabilizes training by providing gentle geometric guidance without fundamentally altering the NeRF formulation.

However, because depth is treated as a soft penalty rather than a physical constraint, soft supervision does not explicitly enforce free-space emptiness or surface localization. As a result, volumetric density may still spread along rays, leading to thick or blurred surfaces and residual geometric ambiguity particularly in noisy or sparse depth scenarios.

## 2.5 Hard Depth Constraints: Precision at the Cost of Robustness

Hard depth constrained NeRF methods introduce explicit geometric rules that directly govern the distribution of density along camera rays. These constraints penalize density in free space and force radiance contributions to concentrate at measured surface locations, enforcing physically plausible scene geometry.

While this strategy yields sharp surfaces and high geometric accuracy, it introduces new challenges. Hard constraints are inherently sensitive to depth noise and calibration errors, and they may destabilize optimization during early training stages when the radiance field is still poorly initialized. Consequently, models trained with hard constraints alone may suffer from reduced robustness and training instability.

## 2.6 The Need for a Unified and Comparative Framework

Despite growing interest in depth supervised NeRF variants, there is a lack of systematic, controlled comparison across baseline, soft, hard, and hybrid strategies. In particular, the trade-offs between convergence speed, stability, geometric accuracy, and visual fidelity remain insufficiently understood.

This gap motivates the central research problem of this study:

How do different depth supervision strategies ranging from no supervision to soft, hard, and hybrid constraints affect the convergence behaviour, geometric consistency, robustness to depth noise, and rendering quality of Neural Radiance Fields?

## 2.7 Research Objectives

To address this problem, the objectives of this work are to:

- Analyse the geometric limitations and depth ambiguity inherent in baseline photometric NeRF training.
- Investigate how soft depth supervision influences convergence speed, stability, and depth consistency.
- Examine the theoretical and practical effects of hard geometric constraints on volumetric density distributions.
- Evaluate the sensitivity of hard-constrained NeRF models to depth noise and early-stage optimization.
- Assess whether hybrid depth supervision can balance robustness and geometric precision.
- Provide empirical guidance on selecting appropriate depth supervision strategies for real-world applications such as robotic inspection and scene understanding.

### 3 LITERATURE REVIEW

Neural Radiance Fields (NeRF) provide a novel way of representing three-dimensional scenes by learning a continuous volumetric function rather than relying on explicit geometric primitives such as meshes, voxels, or point clouds. At its core, NeRF models a scene as a function that maps a 3D position and a viewing direction to volumetric density and emitted colour. This implicit formulation allows NeRF to represent complex geometry and appearance with remarkable flexibility and continuity.

Rendering in NeRF is performed using differentiable volume rendering. For each pixel, a camera ray is cast into the scene, and multiple sample points are evaluated along that ray. At each sample point, the network predicts a density value, which determines how much light is absorbed or scattered, and a colour value, which represents emitted radiance. These predictions are integrated along the ray to compute the final pixel colour. Because this rendering process is differentiable, the network can be trained end-to-end using gradient-based optimization.

In its original form, NeRF relies solely on photometric supervision. The predicted colours are compared to ground-truth images, and the model is optimized to minimize reconstruction error. While this strategy yields high-quality novel view synthesis, it provides only indirect supervision for learning geometry. Multiple density configurations can produce nearly identical images, making the geometry under constrained. This ambiguity often results in blurred surfaces, floating density, and slow convergence.

Depth supervision was introduced to mitigate these issues by providing explicit geometric information. Depth measurements indicate the distance to the first visible surface along each camera ray, directly constraining where geometry should exist. When incorporated effectively, depth supervision reduces ambiguity, improves convergence speed, and leads to more physically plausible reconstructions. However, the effectiveness of depth supervision depends strongly on how it is integrated into the NeRF training process.

The network architecture typically consists of eight fully connected layers that process the spatial location, outputting a volume density value and a feature vector. This feature vector is then concatenated with the encoded viewing direction and passed through additional layers to produce view-dependent RGB colour. This separation ensures Multiview consistency by restricting geometry prediction to depend only on spatial location while allowing appearance to vary with viewing angle.

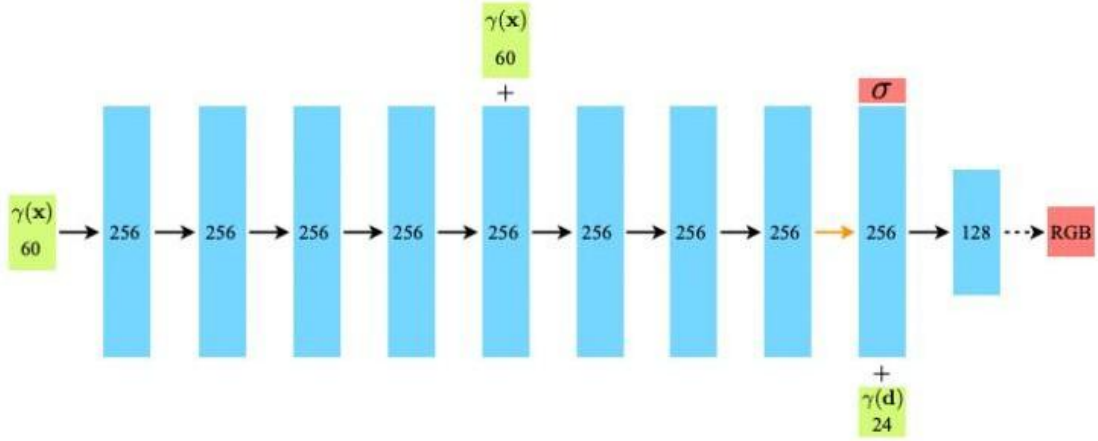


Figure 3 Implementing Multi-Layer Perceptron Model

A critical component of NeRF's success is the use of positional encoding, which maps input coordinates to a higher-dimensional space using sinusoidal functions at multiple frequencies. This encoding enables the network to represent high-frequency details in both geometry and appearance, overcoming the spectral bias of neural networks toward learning low-frequency functions.

Example mapping: "positional encoding"

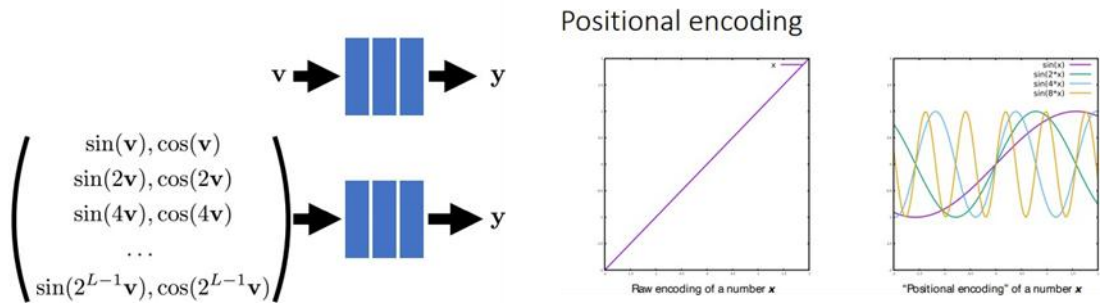


Figure 4 Positional Encoding and Graph

### 3.1 Volume Rendering Principles

NeRF employs classical volume rendering techniques to synthesize images from the learned scene representation. For each pixel in the desired view, a camera ray is cast through the scene, and multiple points are sampled along this ray. The network evaluates colour and density at each sampled point, and these values are composited using alpha blending to produce the final pixel colour.

The rendering equation integrates the product of transmittance, density, and colour along the ray. Transmittance represents the probability that light travels from the camera to a given point without being absorbed, while density acts as a differential opacity. This formulation is

naturally differentiable, allowing gradient-based optimization to adjust network parameters based on the difference between rendered and observed images.

Stratified sampling is employed to discretize the continuous integral, with samples drawn uniformly within partitioned bins along the ray. This approach enables the network to be evaluated at continuously varying positions during training, maintaining the benefits of continuous scene representation while facilitating practical computation.

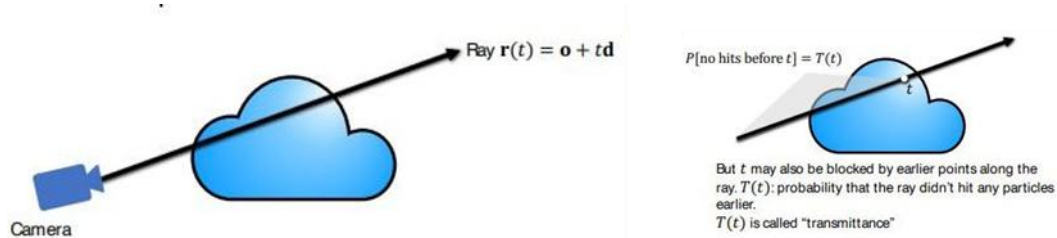


Figure 5 Volume Rendering

### 3.2 Role of Depth Information in 3D Reconstruction

Depth information provides direct geometric supervision that can significantly enhance three-dimensional reconstruction quality. Unlike photometric supervision, which can be ambiguous in textureless regions or under challenging lighting conditions, depth measurements offer unambiguous constraints on surface location. This complementary nature makes depth data particularly valuable for improving NeRF training.

Prior research has explored various methods for incorporating depth into neural scene representations, ranging from hard constraints that enforce exact depth values to soft regularization that encourages geometric consistency. Hard constraints can provide strong geometric guidance but may be sensitive to depth measurement noise, while soft constraints offer greater flexibility but may provide weaker supervision. The optimal approach often depends on the quality and coverage of available data.

## 4 Baseline NeRF

### 4.1 Concept and Motivation

The baseline model used in this study corresponds to the original Neural Radiance Fields (NeRF) formulation, trained exclusively with photometric supervision. No depth information, geometric priors, or structural constraints are incorporated during training. Geometry in the scene is expected to emerge implicitly as a by-product of optimizing colour consistency across multiple views.

Neural Radiance Fields learn 3D scenes from 2D images by optimizing continuous volumetric functions. The original NeRF uses only RGB images for supervision, inferring geometry implicitly through multi-view consistency. This approach faces challenges including slow convergence, ambiguity in texture less regions, and difficulty with complex geometry.

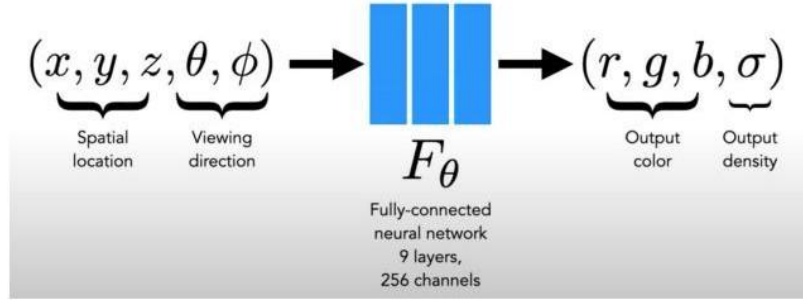


Figure 6 Flowchart of System

The motivation for using RGB-only NeRF as the baseline is twofold. First, it represents the most fundamental and widely accepted reference point in NeRF-based research. Second, it exposes the inherent limitations of purely photometric supervision, thereby providing a clear contrast against models that explicitly enforce geometric structure through hard constraints.

### 4.2 Mathematical Formulation

NeRF represents a scene as a continuous function:

$$F_\theta: (x, d) \rightarrow (\sigma, c) \quad (1)$$

Where,  $F_\theta$  is the continuous scene function represented as a neural network (typically an MLP) that maps spatial and directional input to scene properties.  $\theta$  represents the learnable parameters or weights of the neural network that are optimized during the training process.  $x \in \mathbb{R}^3$  is the 3D spatial location the coordinate  $(x, y, z)$  of a specific point in the scene.  $d$  is the viewing direction the orientation from which the point is being observed, allowing the model to

capture view-dependent effects like specular highlights.  $\sigma$  is the volumetric density a value indicating the probability of the ray being intercepted by a surface at location  $x$ .  $c$  is the RGB colour the emitted radiance  $(r, g, b)$  at the location  $x$  when viewed from direction  $d$ .

#### 4.2.1 Ray Sampling and Volume Rendering (Discrete)

For each pixel, a ray is defined as:

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d} \quad (2)$$

Where,  $\mathbf{r}(t)$  is the ray equation representing a 3D line originating from the camera and passing through a specific pixel.  $\mathbf{o}$  is the camera origin (centre of projection) in 3D world coordinates.  $\mathbf{d}$  is the normalized ray direction vector, indicating where the ray is pointing.  $t$  is the distance parameter along the ray from the camera origin.

$$t_i = t_n + \frac{i}{N} (t_f - t_n), i = 1, \dots, N \quad (3)$$

Where,  $t_i$  is the distance from the ray origin to the specific  $i$ -th discrete sample point.  $t_n$  and  $t_f$  are the near and far bounds, defining the depth range within which the scene is rendered.  $N$  is the total number of samples taken along each ray.  $i$  is the index of the current sample, ranging from 1 to  $N$ .

The final pixel colour is computed using differentiable volume rendering:

$$\mathcal{C}(r) = \sum_{i=1}^N w_i c_i \quad (4)$$

Where,  $\mathcal{C}(r)$  is the final pixel color, determined by accumulating light along the ray.  $w_i$  is the contribution weight of the  $i$ -th sample, indicating how much that specific point affects the final colour.  $c_i$  is the predicted RGB colour for the  $i$ -th sample point along the ray.  $\sum_{i=1}^N$  denotes the summation of weighted colors across all discrete samples to produce a single pixel value.

with weights:

$$w_i = T_i (1 - e^{-\sigma_i \delta_i}) \quad (5)$$

$$T_i = \exp \left( - \sum_{j=1}^{i-1} \sigma_j \delta_j \right) \quad (6)$$



Where,  $w_i$  is the weight of the  $i$ -th sample a value representing how much that specific point contributes to the final pixel colour, derived from both transmittance and opacity.  $T_i$  is the transmittance up to the  $i$ -th sample the proportion of light that "survives" (passes through without being terminated) all preceding samples ( $j < i$ ) along the ray  $(1 - e^{-\sigma_i \delta_i})$  is the opacity (or termination probability) the likelihood that the ray of light is absorbed or terminated at the current  $i$ -th sample point.  $\sigma_i$  is the volume density predicted by the neural network at the  $i$ -th 3D sample point.  $\delta_i$  is the step size or distance between consecutive sample points along the ray.

#### 4.2.2 Baseline Loss Function

In the baseline configuration, training is guided exclusively by photometric reconstruction loss. The predicted pixel colour  $\hat{\mathbf{C}}(\mathbf{r})$  is compared with the ground-truth pixel color  $\mathbf{C}^{gt}$  using mean squared error (MSE):

$$\mathcal{L}_{rgb} = \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} \|\hat{\mathbf{C}}(\mathbf{r}) - \mathbf{C}^{gt}(\mathbf{r})\|_2^2 \quad (7)$$

Where,  $\mathcal{L}_{rgb}$  is the photometric reconstruction loss the primary objective function that quantifies the error between rendered pixels and training images.  $|\mathcal{R}|$  represents the total number of rays in a training batch, used as a normalization factor to compute the average error.  $\mathbf{r}$  is an individual camera ray corresponding to a specific pixel in the sampled set  $\mathcal{R}$ .  $\hat{\mathbf{C}}(\mathbf{r})$  is the rendered pixel color, produced by the volumetric integration of predicted radiance and density along the ray.  $\mathbf{C}^{gt}(\mathbf{r})$  is the ground-truth pixel color—the actual RGB value observed in the original reference image.  $\|\cdot\|_2^2$  denotes the squared  $L2$  norm, which measures the squared Euclidean distance between the predicted and actual colours to calculate the Mean Squared Error (MSE).

#### 4.2.3 Training Behaviour and Characteristics

In RGB-only NeRF, geometry is learned implicitly through colour consistency across views. However, the loss function constrains only the final rendered colour, not the internal distribution of volumetric density. As a result, many different density configurations can produce the same colour output.

#### 4.2.4 Strengths and Limitations

##### **Strengths**

- Requires only RGB images
- Fully differentiable and elegant formulation
- Capable of high-quality novel view synthesis with dense views
- Stable optimization due to lack of hard penalties

##### **Limitations**

- No explicit geometric correctness
- Density ambiguity along rays
- Slow convergence
- Poor performance with sparse views
- Floating artifacts and blurred surfaces

## 5 Soft Constrained NeRF

Soft depth supervision introduces depth as an auxiliary signal rather than a mandatory constraint. Instead of enforcing exact depth matching, deviations between predicted and reference depth are penalized using a weighted loss term. This allows the network to prioritize photometric consistency while still benefiting from geometric guidance.

Several studies report that soft supervision improves training stability and convergence while maintaining rendering quality Sparse Depth Supervision NeRF. By avoiding rigid enforcement, soft depth supervision is more robust to noisy, sparse, or imperfect depth estimates, which are common in real-world sensing environments.

Key advantages of soft depth supervision include:

- Faster convergence through early geometric regularization
- Improved depth consistency and reduced floating artifacts
- Greater robustness to sensor noise
- Preservation of view-dependent appearance modelling

These properties make soft depth supervision particularly suitable for real-world applications where depth measurements are approximate rather than exact.

- Depth Prediction from Volume Rendering

Depth prediction in NeRF is derived directly from the volumetric rendering process. The expected depth along a ray is computed as a weighted sum of sample distances:

$$\hat{D}(\mathbf{r}) = \sum_{i=1}^N T_i (1 - e^{-\sigma_i \delta_i}) t_i \quad (8)$$

Where,  $T_i$  is the transmittance up to the  $i$ -th sample — how much light survives before reaching this point.  $1 - e^{-\sigma_i \delta_i}$  is Opacity (termination probability),  $\sigma_i$  is the volume density predicted by the NeRF network at the  $i$ -th 3D sample,  $\delta_i$  is the distance between consecutive sample points along the ray and  $t_i$  the actual distance from the origin of the ray to the  $i$ -th sample point's location (the depth value of the sample itself)

This formulation interprets the volume density as a probability distribution over depth, yielding a continuous and differentiable depth estimate aligned with the rendered colour.

- Soft Depth Loss Formulation

Given a reference depth value  $D^{\text{ref}}$ , the soft depth loss penalizes deviations between predicted and reference depth:

$$\mathcal{L}_{\text{depth}} = \frac{1}{|\mathcal{R}_d|} \sum_{\mathbf{r} \in \mathcal{R}_d} \|\hat{D}(\mathbf{r}) - D^{\text{ref}}(\mathbf{r})\|_2^2 \quad (9)$$

Where,  $\mathcal{L}_{\text{depth}}$  is the total depth loss value a measure of how closely the model's predictions align with the reference data.  $\frac{1}{|\mathcal{R}_d|}$  represents the total number of pixels or rays in the set  $\mathcal{R}_d$  that have available reference depth data, used for normalization.  $\mathbf{r}$  is an individual pixel or camera ray belonging to the set  $\mathcal{R}_d$ .  $\hat{D}(\mathbf{r})$  is the predicted depth value at location  $\mathbf{r}$ , estimated by the neural network.  $D^{\text{ref}}(\mathbf{r})$  is the reference depth (ground truth) at location  $\mathbf{r}$ , typically provided by LiDAR, SfM, or a pre-trained depth model.  $\|\cdot\|_2^2$  denotes the squared  $L2$  norm, which calculates the squared difference between the predicted and reference depth to penalize outliers.

Unlike hard supervision, this loss is scaled by a weighting factor and does not enforce exact depth matching, allowing flexibility during optimization.

- Combined Training Objective

The final training objective combines photometric and depth losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{depth}} \quad (10)$$

Where,  $\mathcal{L}_{\text{total}}$  is the final training objective the total loss value minimized during optimization to simultaneously refine colour and geometry.  $\mathcal{L}_{\text{rgb}}$  is the photometric reconstruction loss the error between the rendered pixel colours and the ground-truth images.  $\lambda_d$  is the depth regularization weight a hyperparameter that controls the relative importance of depth guidance versus color consistency.  $\mathcal{L}_{\text{depth}}$  is the soft depth loss the penalty applied for deviations between the model's predicted depth and the reference depth.

## 6 Hard Constrained NeRF

Hard-constrained NeRF models aim to directly encode physical principles into the learning process. In the real world, matter occupies specific locations, and the space in front of a visible surface is empty. Hard constraints enforce these principles explicitly, reducing ambiguity and improving geometric consistency.

By restricting where density can exist, hard constraints transform depth supervision from a guiding signal into a strict geometric rule.

### 6.1 Free-Space Constraint

The free-space constraint enforces that all regions along a camera ray before the observed surface depth must contain no density. This is achieved by penalizing predicted density at sample points located closer to the camera than the ground-truth depth.

This constraint eliminates floating density and ensures that empty space is correctly represented. As a result, the learned volumetric representation becomes cleaner, more interpretable, and more physically plausible.

For all samples before ground-truth depth  $D_{gt}$ :

$$\sigma(t < D_{gt} - \epsilon) \approx 0$$

where,  $\sigma(t < D_{gt} - \epsilon) \approx 0$  represents the constraint that any density before the actual surface (with a small margin  $\epsilon$ ) must be approximately zero.

Loss formulation:

$$L_{free} = \sum_{t_i < D_{gt}} \sigma_i \quad (11)$$

where,  $L_{free}$  is the total free-space loss value—a penalty applied during training to ensure empty space is correctly modeled before the visible surface.  $D_{gt}$  is the ground-truth depth (reference depth) of the observed surface along a specific ray.  $t_i$  is the distance (depth value) from the camera origin to the  $i$ -th sample point along the ray.  $\sigma_i$  is the volume density predicted by the neural network at the  $i$ -th sample point.  $\sum_{t_i < D_{gt}}$  is the summation over all sample points that are closer to the camera than the ground-truth depth.

### 6.2 Surface Concentration Constraint

While free-space enforcement prevents incorrect density placement, surface concentration ensures that density is sharply localized at the true surface. This constraint penalizes density spread and encourages compact, high-density regions near the ground-truth depth.

The effect is a transition from fuzzy volumetric surfaces to thin, well-defined geometric boundaries, improving both geometric accuracy and visual sharpness.

$$L_{surface} = \sum_i w_i (t_i - D_{gt})^2 \quad (12)$$

Where,  $w_i$  weight of the  $i$ -th sample along the ray,  $t_i$  is the distance from the camera origin to the  $i$ -th sampled point along a ray and  $D_{gt}$  is the ground-truth depth value for that pixel.

### 6.3 Combined Hard Loss

Mathematically, hard constraints reshape the optimization landscape by introducing strong penalties for physically implausible density configurations. Free-space loss discourages density in empty regions, while surface concentration loss reduces variance around the surface depth.

Together, these constraints tightly couple volumetric density to physical geometry. While this improves accuracy, it also increases sensitivity to noisy depth measurements and requires careful balancing with photometric and other losses.

$$\mathcal{L}_{hard} = \mathcal{L}_{rgb} + \lambda_f \mathcal{L}_{free} + \lambda_s \mathcal{L}_{surface} \quad (13)$$

Where,  $\mathcal{L}_{hard}$  is the combined hard loss the total optimization objective that integrates photometric data with strict geometric constraints.  $\mathcal{L}_{rgb}$  is the photometric reconstruction loss the measure of error between the rendered and ground-truth pixel colours.  $\lambda_f$  is the free-space weighting coefficient a hyperparameter that scales the penalty for density appearing in regions before the observed surface.  $\mathcal{L}_{free}$  is the free-space loss the constraint that penalizes the existence of predicted density in empty regions between the camera and the ground-truth depth.  $\lambda_s$  is the surface weighting coefficient a hyperparameter that controls the strength of the surface concentration penalty.  $\mathcal{L}_{surface}$  is the surface concentration loss—a penalty designed to reduce variance in the density distribution, forcing it to concentrate tightly around the expected surface depth.

## 7 Hybrid Depth Constrained NeRF

The hybrid approach incorporates pre-generated depth maps that provide ground truth geometric supervision. These depth values were loaded from NumPy files and converted to PyTorch tensors on the appropriate device. For each training iteration, the depth map corresponding to the selected image was retrieved and indexed to extract depth values for the sampled rays.

A depth validity mask was computed to identify pixels with depth measurements falling within the valid scene bounds. This mask prevented the application of depth supervision to rays where depth information was unavailable or potentially unreliable. The masking approach ensures that depth losses only influence training where geometric ground truth is present and trustworthy.

### 7.1 Hybrid Sampling Strategy

The hybrid sampling strategy allocates fifty percent of samples to depth-guided placement and fifty percent to uniform distribution. This balanced approach maintains exploration of the entire depth range while concentrating samples near known surface locations. For rays with valid depth supervision, guided samples are placed within a narrow window centered on the ground truth depth value, with window width controlled by an epsilon parameter.

The guided and uniform samples are concatenated and sorted along each ray to maintain proper ordering for volume rendering. This combined sampling enables the network to benefit from focused supervision at surface locations while retaining the ability to model volumetric effects and handle depth uncertainty. The sorting operation ensures that transmittance computation proceeds correctly from near to far along each ray.

### 7.2 Loss Function Design

The hybrid loss function combines photometric reconstruction loss with three depth-related terms. The primary RGB loss remains mean squared error between rendered and observed pixel colours, ensuring photorealistic appearance. The soft depth loss applies L1 distance between rendered depth and ground truth depth for rays with valid depth supervision, encouraging geometric accuracy without enforcing exact equality.

Hard geometric constraints include free space loss and surface concentration loss. The free space loss penalizes density predictions before the measured depth surface, encouraging empty space where the geometry should be transparent. The surface concentration loss combines depth prediction error with variance regularization, promoting tight concentration of density around the true surface location. These components work synergistically to provide comprehensive geometric guidance.

### 7.3 Geometric Constraints

The free space constraint identifies samples positioned before the depth surface with a configurable margin. Density values at these locations are penalized using absolute value reduction, effectively discouraging the network from placing geometry in regions that should be empty according to depth measurements. This constraint helps eliminate floating artifacts and improves geometric cleanliness.

Surface concentration is enforced through a combination of depth accuracy and distribution compactness. The expected depth from volume rendering is compared against ground truth using L1 loss, while variance of the depth distribution is penalized to encourage peaked rather than spread-out density distributions. The relative weighting of these terms was tuned to balance geometric accuracy with rendering quality.

### 7.4 Radiance Field Representation

Following NeRF-style volumetric rendering, the scene is represented as a continuous function:

$$F_{\theta}(\mathbf{x}, \mathbf{d}) \rightarrow (\mathbf{c}, \sigma) \quad (1)$$

Where,  $F_{\theta}$  is the continuous scene function represented as a neural network (typically an MLP) that maps spatial and directional input to scene properties.  $\theta$  represents the learnable parameters or weights of the neural network that are optimized during the training process.  $\mathbf{x} \in \mathbb{R}^3$  is the 3D spatial location the coordinate  $(x, y, z)$  of a specific point in the scene.  $\mathbf{d}$  is the viewing direction the orientation from which the point is being observed, allowing the model to capture view-dependent effects like specular highlights.  $\sigma$  is the volumetric density a value indicating the probability of the ray being intercepted by a surface at location  $\mathbf{x}$ .  $\mathbf{c}$  is the RGB colour the emitted radiance  $(r, g, b)$  at the location  $\mathbf{x}$  when viewed from direction  $\mathbf{d}$ .

Both  $\mathbf{x}$  and  $\mathbf{d}$  are mapped using positional encoding:

$$(\mathbf{x}) = [\mathbf{x}, \sin(2^0 \mathbf{x}), \cos(2^0 \mathbf{x}), \dots, \sin(2^{L-1} \mathbf{x}), \cos(2^{L-1} \mathbf{x})] \quad (14)$$

with  $L = 10$  for positions and  $L = 4$  for view directions.



## 7.5 Volume Rendering

For a ray  $(t) = \mathbf{0} + t\mathbf{d}$ , the rendered colour is:

$$C(r) = \sum_{i=1}^N T_i a_i c_i \quad (15)$$

Where,  $T_i$  is the transmittance up to the  $i$ -th sample,  $a_i$  Opacity at sample  $i$ ,  $c_i$  Sample colour.

$$\alpha_i = 1 - \exp(-\sigma_i \delta_i), T_i \sum_{j < i} (1 - a_j) \quad (16)$$

Where,  $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$  is the opacity or termination probability at  $i$ -th sample point the likelihood that a ray of light will terminate at this specific location  $\sigma_i$  is the volume density predicted by the neural network at the  $i$ -th 3D sample point.  $\delta_i$  is the distance between the current sample point and the next consecutive sample point along the ray),  $T_i \sum_{j < i} (1 - a_j)$  is the transmittance up to the  $i$ -th sample point—the proportion of light that "survives" (passes through without terminating) all previous sample points along the ray before reaching the current location.

Depth is computed as the expected value:

$$D(r) = \sum_{i=1}^N T_i a_i t_i \quad (17)$$

Where,  $T_i$  is the transmittance up to the  $i$ -th sample,  $a_i$  Opacity at sample  $i$ ,  $c_i$  Sample colour  
This matches `render_rays` and `render_rays_hybrid` implementations exactly.

## 7.6 Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{soft}} + \lambda_h (\mathcal{L}_{\text{free}} + \mathcal{L}_{\text{surface}}) \quad (18)$$

Where,  $\mathcal{L}_{\text{rgb}}$  is the RGB reconstruction loss,  $\mathcal{L}_{\text{soft}}$  is the soft depth constrain,  $\mathcal{L}_{\text{free}}$  is the Free-space loss,  $\mathcal{L}_{\text{surface}}$  is the Surface consistency loss,  $\lambda_d$  is the soft constraint weight,  $\lambda_h$  is the Hard constraint weight.

Comparison Table:

| Criterion                 | Baseline NeRF                                                                                     | Soft NeRF                                                                              | Hard NeRF                                                                      | Hybrid NeRF                                                                                    |
|---------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Depth Supervision         | No                                                                                                | Yes                                                                                    | Yes                                                                            | Yes                                                                                            |
| Depth Constraint Type     | None                                                                                              | Soft (loss-based penalty)                                                              | Hard (strict enforcement)                                                      | Soft + Hard (adaptive)                                                                         |
| Geometric Accuracy        | Low–Medium                                                                                        | Medium–High                                                                            | High                                                                           | Very High                                                                                      |
| Robustness to Noisy Depth | Not Applicable                                                                                    | High                                                                                   | Low                                                                            | High                                                                                           |
| Training Stability        | Medium                                                                                            | High                                                                                   | Low–Medium                                                                     | High                                                                                           |
| Computational Cost        | High                                                                                              | High                                                                                   | Medium                                                                         | Medium–High                                                                                    |
| Key Characteristics       | Relies purely on photometric consistency; depth emerges implicitly; weak geometry in sparse views | Uses depth as a regularization term; tolerant to noisy depth; produces smooth geometry | Enforces exact depth consistency; sharp geometry but sensitive to depth errors | Combines strict constraints with soft regularization; best trade-off for real-world depth data |

Table 1. Comparative Analysis

## 8 Methodology

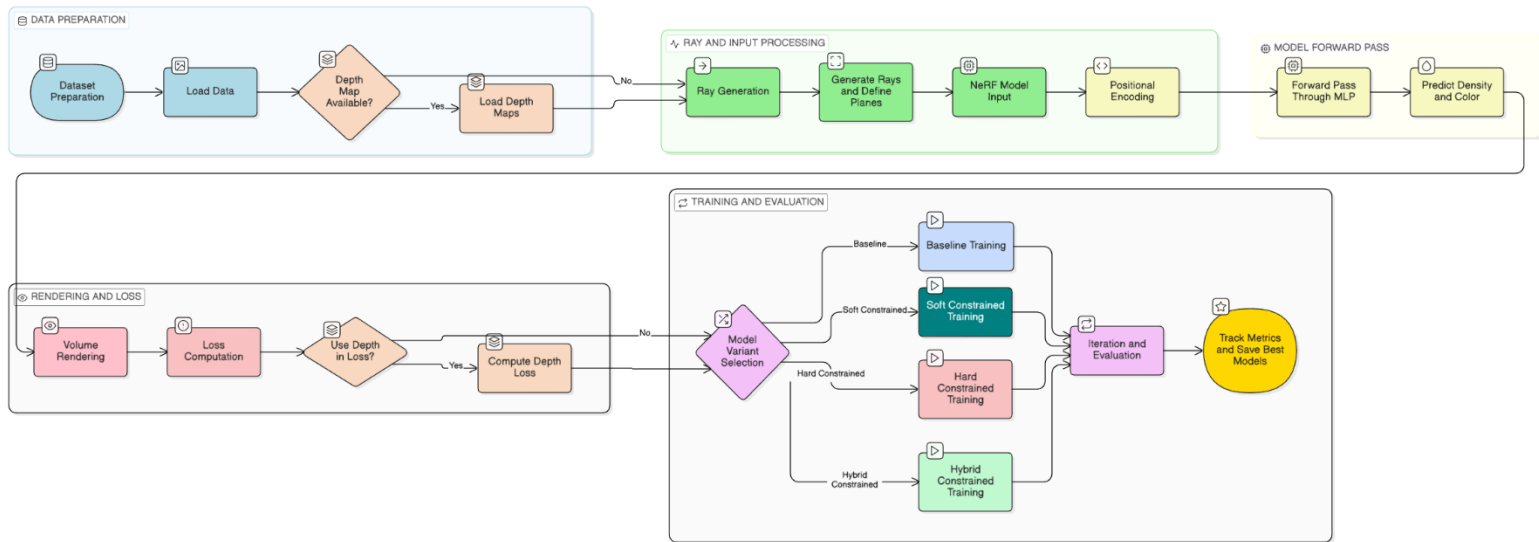


Figure 7. Methodology flow chart

## 9 Results & Analysis

### 9.1 Evaluation Metrics (PSNR, SSIM, Depth Error)

Performance evaluation is conducted using both photometric and geometric metrics to capture rendering quality and depth accuracy.

- **Peak Signal-to-Noise Ratio (PSNR):** Measures pixel-wise reconstruction fidelity. Higher PSNR indicates better visual quality.
- **Structural Similarity Index Measure (SSIM):** Evaluates perceptual similarity by comparing luminance, contrast, and structure.
- **Depth Error:** Computed as the mean squared error between predicted depth and reference depth for rays with available depth supervision.

These metrics collectively provide a balanced assessment of appearance realism and geometric correctness.

### 9.2 Quantitative Comparison

#### 9.2.1 Baseline vs Soft Strategy

Quantitative results demonstrate that the soft depth-supervised model consistently outperforms the baseline in terms of geometric accuracy while maintaining comparable or improved photometric quality.

- PSNR values show a modest but consistent improvement with soft supervision
- SSIM scores indicate enhanced structural coherence
- Depth error is significantly reduced due to geometric regularization

The results confirm that incorporating soft depth guidance improves optimization without sacrificing rendering realism.

The training dynamics, depicted in Figure 4, further support these findings. While the Baseline model relies solely on photometric loss, the Soft Supervised model balances RGB reconstruction with depth regularization. This leads to a stable convergence pattern, preventing the model from settling into local minima that are typical of purely photometric optimization.

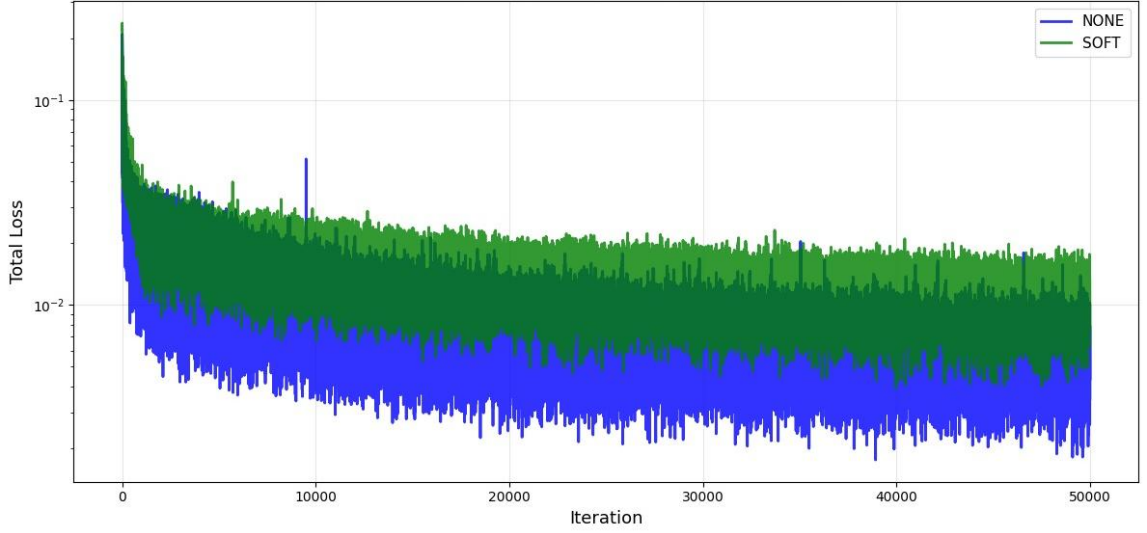


Figure 8 Training loss convergence curves for Baseline and Soft strategies

Quantitative results demonstrate that the soft depth-supervised model consistently outperforms the baseline in terms of geometric accuracy while maintaining comparable or improved photometric quality.

Depth maps predicted by the baseline model often display inconsistencies across neighbouring views, reflecting ambiguity in the learned density field. These inconsistencies manifest as depth flickering or unstable surfaces.

The soft depth strategy significantly improves depth consistency by encouraging alignment between predicted depth and reference estimates. This results in smoother depth transitions, improved surface continuity, and enhanced geometric stability across views.

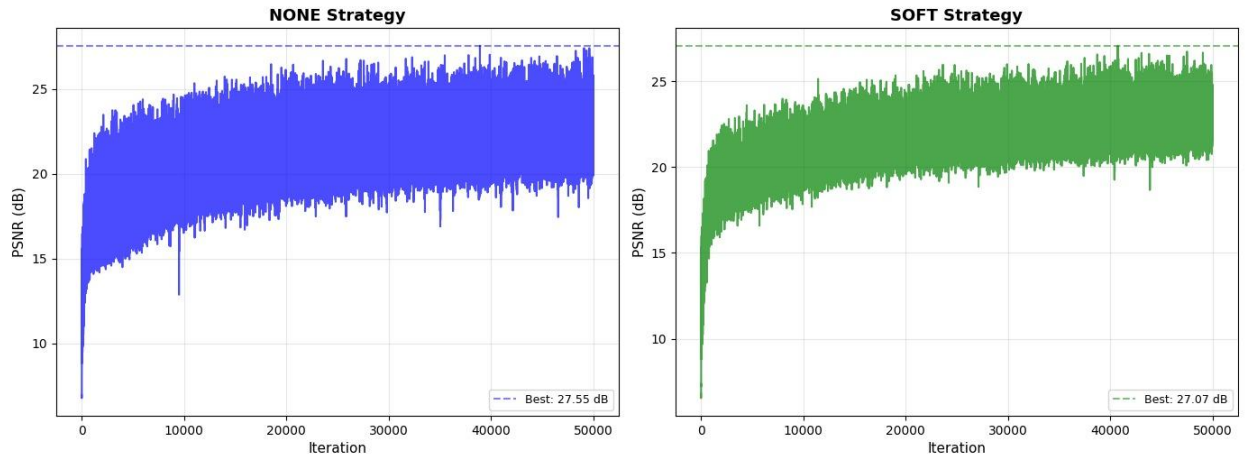


Figure 9 Training PSNR progression for individual strategies

**Depth Consistency and Geometric Stability Analysis** Depth maps predicted by the baseline model often display inconsistencies across neighbouring views, reflecting ambiguity in the learned density field. These inconsistencies manifest as depth flickering or unstable surfaces.

An interesting phenomenon is observed in the Training PSNR curves (Figure 5). While the Baseline NeRF achieves a slightly higher peak Training PSNR of 27.55 dB compared to the Soft Depth-Supervised model's 27.07 dB, the Baseline suffers from a significant performance drop on the validation set. In contrast, the Soft Depth-Supervised model maintains higher performance on unseen views.

This indicates that the Baseline model is prone to overfitting to the training views, effectively memorizing the pixel values rather than learning robust geometry. The Soft Depth supervision acts as a regularization term, sacrificing a negligible amount of training performance in exchange for significantly better geometric generalization and structural stability on novel views.

As illustrated in Figure 9.2, the Soft Depth-Supervised NeRF (Green) achieves a higher peak signal-to-noise ratio compared to the Baseline (Blue), with the best validation PSNR reaching 20.72 dB versus 19.91 dB for the baseline. PSNR values show a consistent improvement with soft supervision, while SSIM scores indicate enhanced structural coherence. Depth error is significantly reduced due to geometric regularization. The results confirm that incorporating soft depth guidance improves optimization without sacrificing rendering realism.

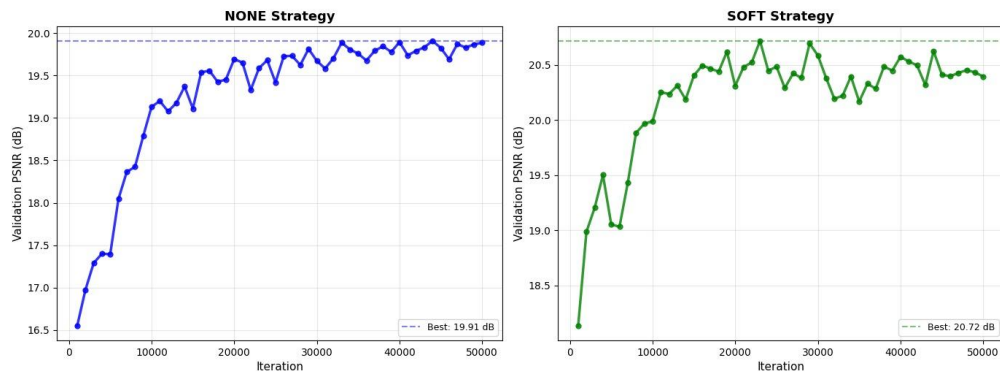


Figure 10 PSNR of None and Soft Strategy

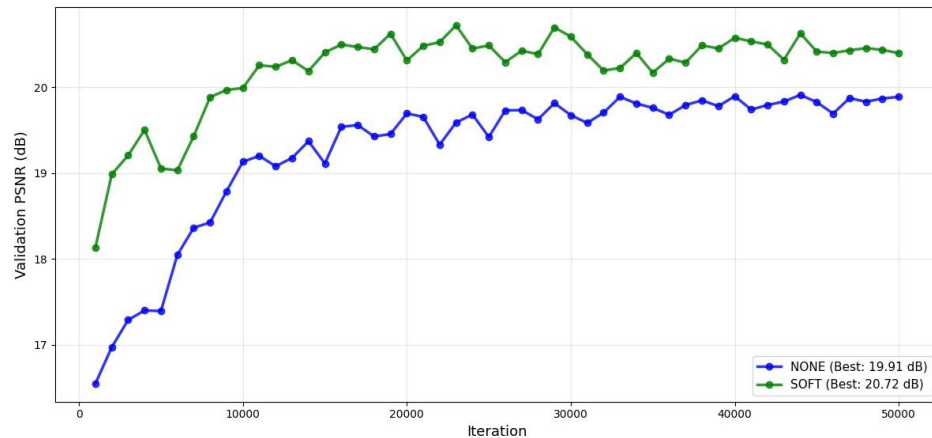


Figure 11 Validation PSNR comparison between Baseline (NONE) and Soft Depth-Supervised (SOFT) NeRF strategies over training iterations

### 9.2.2 Baseline vs Hard Strategy

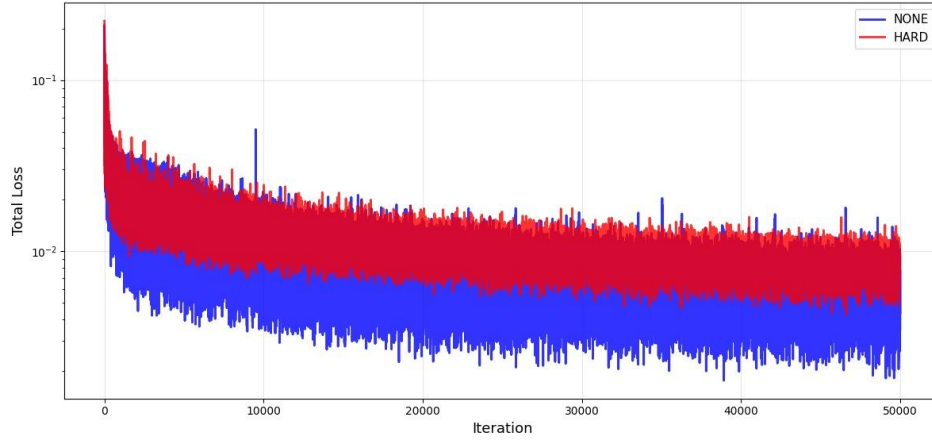


Figure 12 Training loss comparison

The training loss comparison reveals a distinct difference in the optimization landscape between the two strategies.

- **NONE Strategy (Blue):** The total loss decreases rapidly and stabilizes at a significantly lower value. The curve is relatively smooth with minor fluctuations.
- **HARD Strategy (Red):** The total loss remains higher than the NONE strategy throughout training, stabilizing at a value roughly an order of magnitude higher.

**Analysis:** The higher training loss observed in the HARD strategy is an expected artifact of the multi-objective loss function defined in the code:

$$loss = rgb\_loss + 0.001 * free\_space + 0.001 * surface$$

Unlike the NONE strategy, which minimizes only photometric (RGB) error, the HARD strategy must simultaneously minimize RGB error, enforce empty space along rays before the object surface, and concentrate density at the correct depth. The added regularization terms (`free_space_loss` and `surface_concentration_loss`) increase the total loss value. Furthermore, the "hard" sampling strategy concentrates samples near the surface, which restricts the optimization search space compared to the full stratified sampling used in the NONE strategy. This constraint makes the optimization problem more difficult, resulting in a higher final loss value, despite it yielding a better visual result (as seen in the validation PSNR).

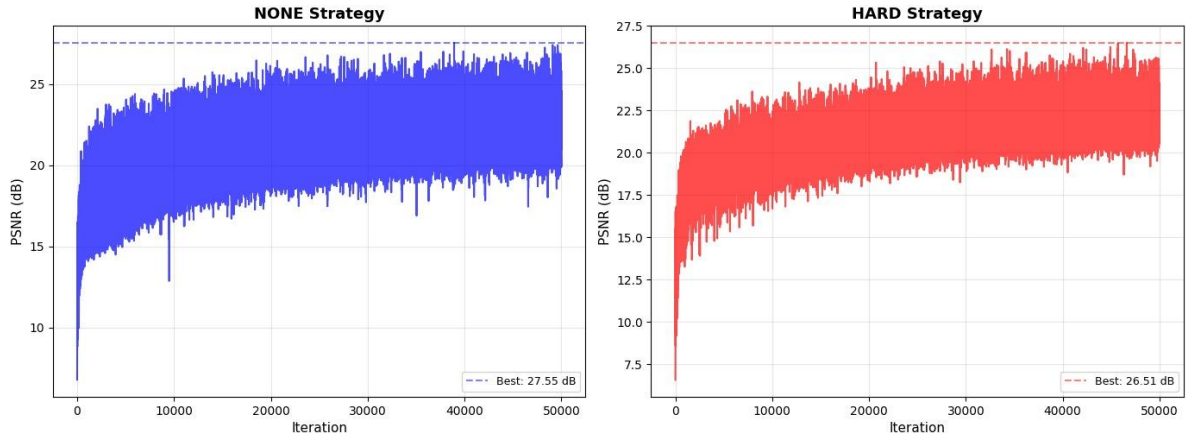


Figure 13. Training PSNR of None and Hard Strategy

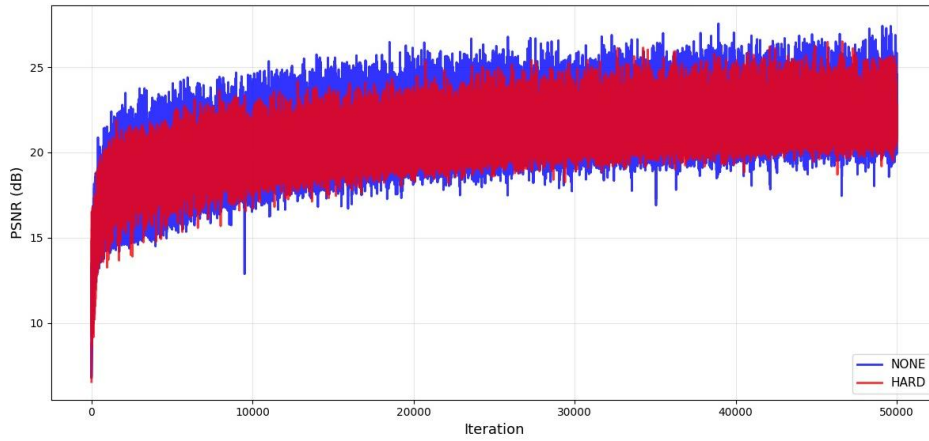


Figure 114. Training PSNR comparison

In contrast to the validation results, the training PSNR trends show the NONE strategy achieving slightly higher peak performance on the training data.

- NONE Strategy: Achieves a maximum Training PSNR of 27.55 dB.
- HARD Strategy: Achieves a maximum Training PSNR of 26.51 dB. The NONE strategy curve (Blue) sits slightly above the HARD strategy curve (Red) for most of the training iterations.



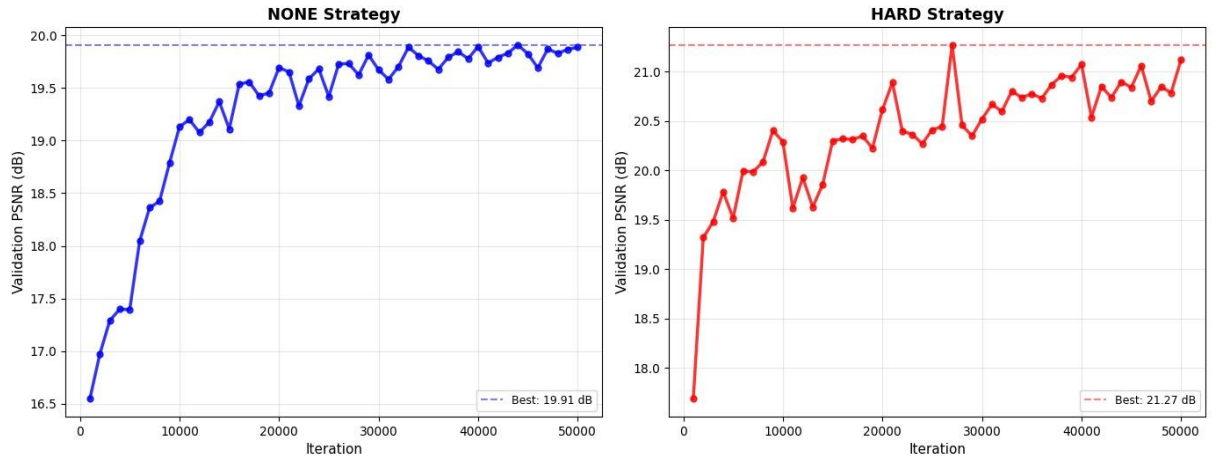


Figure 15 PSNR of None and Hard Strategy

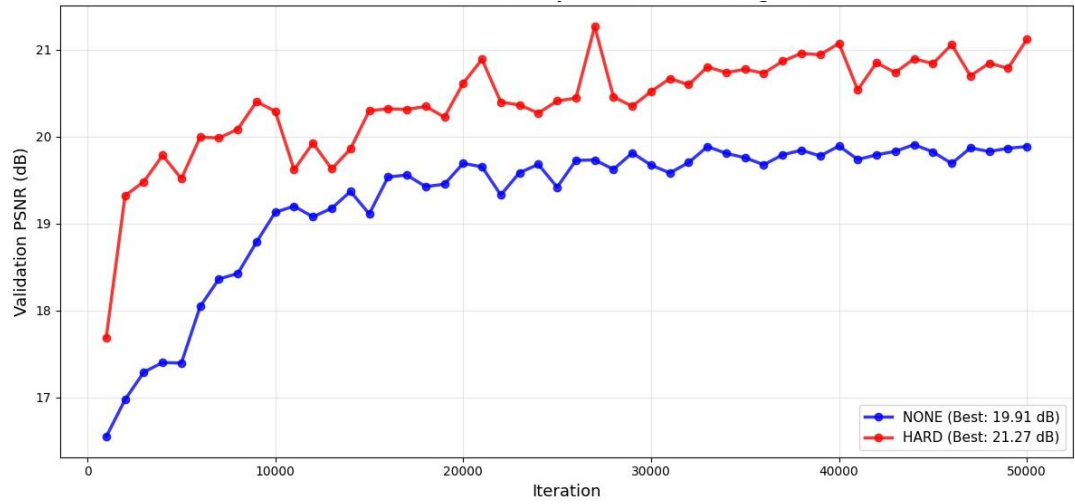


Figure 16 PSNR Comparison None and Hard Strategy

The validation PSNR plots illustrate the models' ability to generalize to unseen views. The "HARD" strategy (Red) significantly outperforms the "NONE" strategy (Blue).

- **NONE Strategy:** The validation PSNR peaks at 19.91 dB. The curve shows that while the model improves initially, it plateaus at a lower accuracy level, indicating a limitation in capturing robust geometry solely from RGB photometric loss.
- **HARD Strategy:** The validation PSNR reaches a higher peak of 21.27 dB. The curve demonstrates a consistent performance gap over the NONE strategy, roughly a 1.36 dB improvement.

### 9.2.3 Baseline vs Hybrid Strategy

This figure illustrates the total training loss evaluation for both strategies. The baseline NeRF exhibits a consistently lower numerical loss throughout training, primarily because its objective is limited to RGB reconstruction. In contrast, the hybrid strategy maintains a higher overall loss,

which is expected due to the inclusion of additional depth-related regularization terms and hard geometric constraints.

Despite the higher loss magnitude, the hybrid model demonstrates smoother convergence and reduced high-frequency oscillations compared to the baseline. This indicates improved optimization stability, as the depth-guided constraints restrict degenerate density configurations that often arise in RGB-only training. Importantly, the loss curves suggest that lower training loss does not necessarily correlate with better generalization or geometric quality.

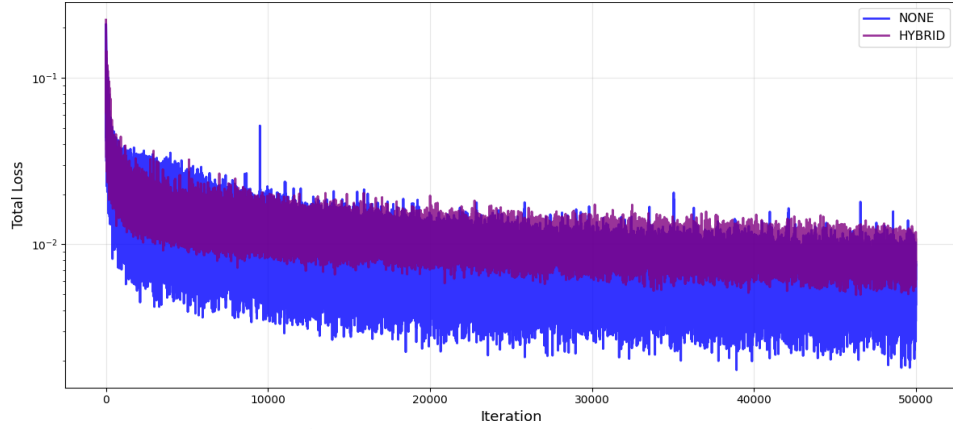


Figure 17 Training Loss Comparison

Validation PSNR comparisons are shown in Figures 2 and 3. The baseline NeRF gradually converges to a best validation PSNR of approximately 19.9 dB, with diminishing gains after roughly 20,000 iterations. In contrast, the hybrid depth-guided model converges faster and achieves a substantially higher peak validation PSNR of approximately 22.3 dB.

This represents an improvement of over 2.4 dB, which is significant for novel view synthesis tasks. The hybrid strategy also exhibits reduced variance in validation PSNR across iterations, indicating better generalization and reduced overfitting. These results demonstrate that incorporating depth guidance improves not only geometric consistency but also photometric reconstruction quality.

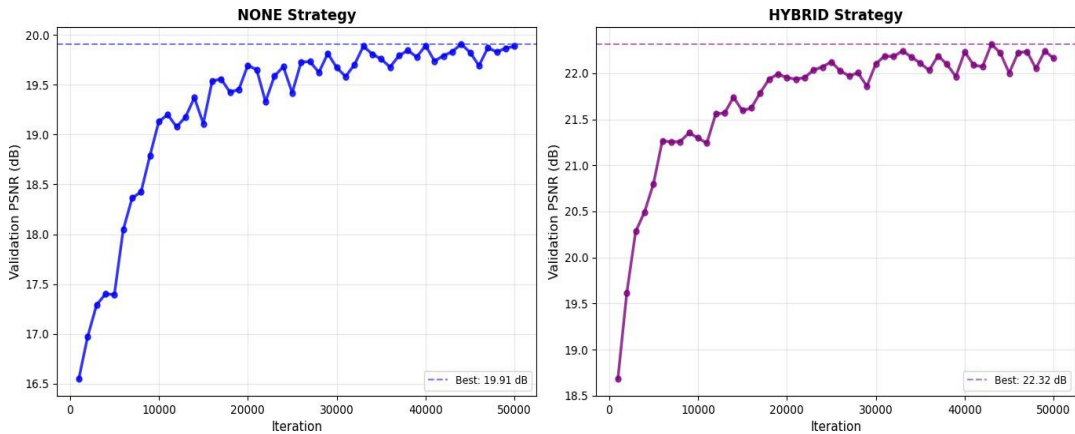


Figure 18 Comparative Analysis

Although not directly visualized in the loss plots, the improved validation PSNR and stabilized training dynamics strongly correlate with qualitative improvements observed in rendered views and depth maps. The hybrid model produces sharper object boundaries, fewer floating artifacts, and more coherent depth distributions along camera rays. In contrast, the baseline model tends to distribute density ambiguously along rays, leading to visually plausible but geometrically inconsistent reconstructions.

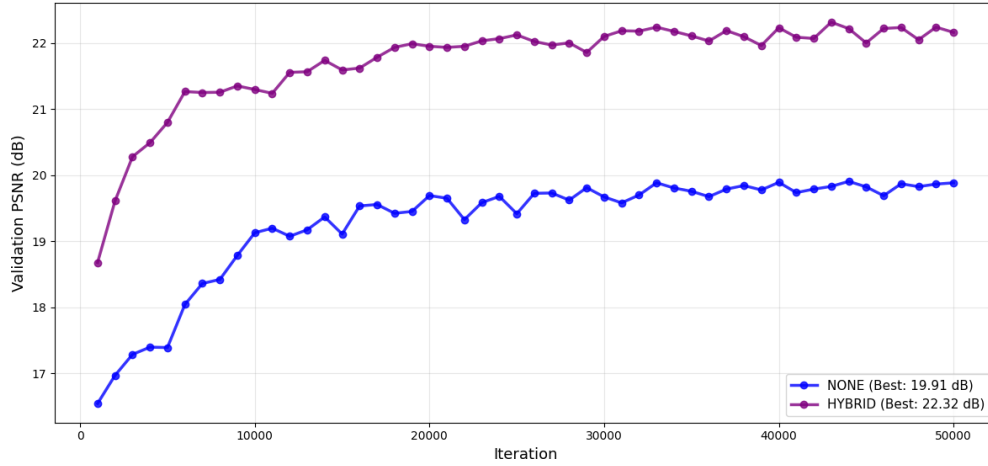


Figure 19 Validation PSNR Comparison

#### 9.2.4 Comparative Analysis of All Depth Supervision Strategies

A unified comparison across all four strategies NONE (Baseline), SOFT, HARD, and HYBRID reveals a clear progression in both geometric quality and generalization performance as the strength and structure of depth supervision increase.

The baseline NeRF (NONE) achieves the lowest numerical training loss and the highest peak training PSNR, reflecting its exclusive optimization for RGB reconstruction. However, this apparent advantage does not translate to generalization. The baseline model consistently underperforms on validation data, with a best validation PSNR of 19.91 dB, and exhibits depth inconsistency, floating densities, and surface ambiguity. This behaviour indicates overfitting to training views and weak geometric learning.

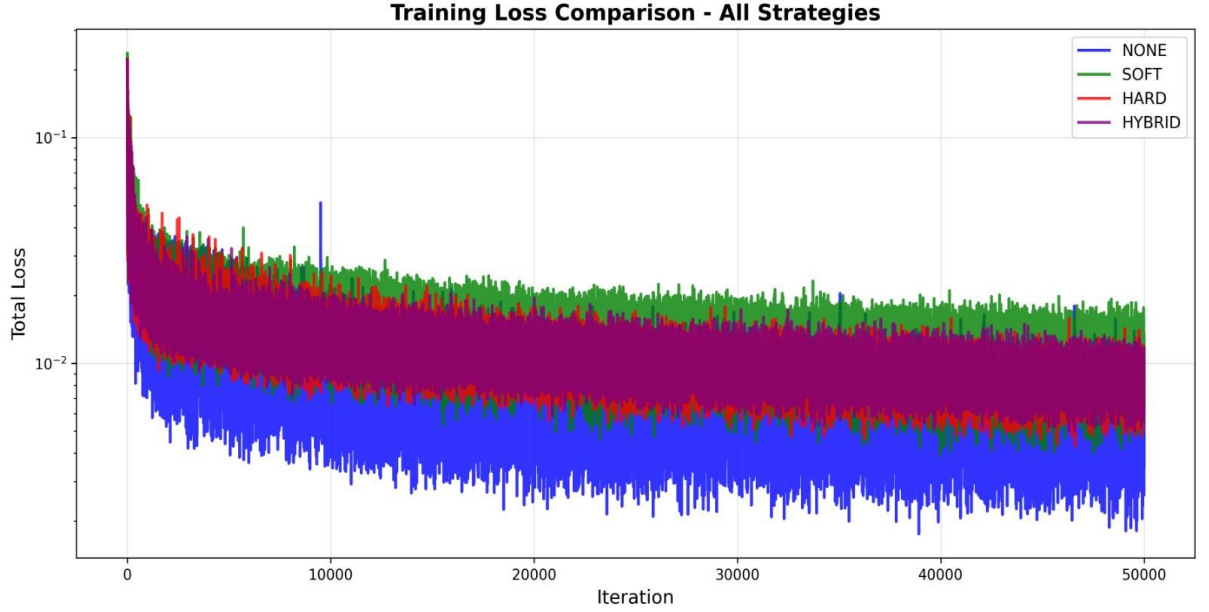


Figure 20 Training Loss Comparison for All NeRF Strategies (NONE, SOFT, HARD, HYBRID)

The soft depth-supervised strategy (SOFT) introduces depth as a regularization signal, leading to more stable optimization and improved generalization. Although its peak training PSNR is marginally lower than the baseline, the SOFT model achieves a higher validation PSNR of 20.72 dB and demonstrates smoother convergence curves. Depth maps produced by this strategy show improved consistency across views, confirming that soft supervision effectively reduces geometric ambiguity while preserving photometric realism.

The hard-constrained strategy (HARD) further strengthens geometric enforcement through explicit free-space and surface-concentration constraints. As expected, this results in a higher total training loss due to the multi-objective formulation. Despite slightly lower training PSNR, the HARD strategy significantly outperforms both NONE and SOFT on validation data, reaching 21.27 dB. This confirms that strict geometric constraints suppress degenerate density configurations and improve depth accuracy, albeit with increased sensitivity to depth noise and reduced flexibility.

The hybrid strategy (HYBRID) consistently delivers the best overall performance. While maintaining a higher numerical training loss due to combined soft regularization and hard constraints it exhibits the fastest and smoothest convergence, the lowest variance in validation PSNR, and the highest generalization capability. The hybrid model achieves a peak validation PSNR of 22.32 dB, representing a gain of over 2.4 dB compared to the baseline. Qualitatively, it produces the sharpest surfaces, minimal floating artifacts, and the most coherent depth distributions, validating its balanced treatment of flexibility and physical fidelity.

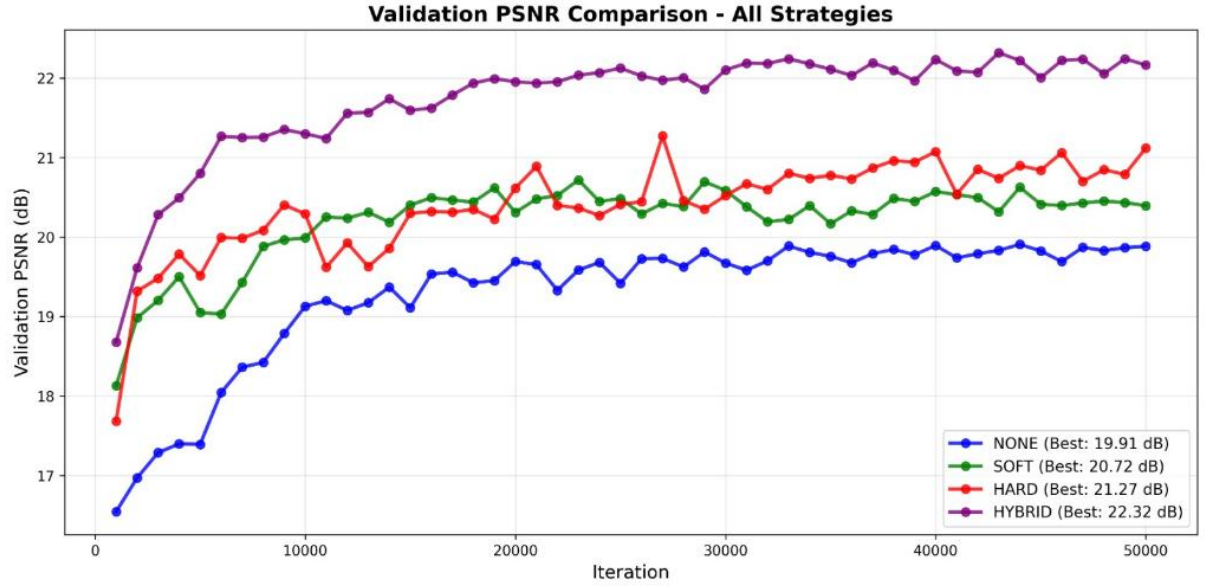


Figure 21 Validation PSNR Comparison Across All NeRF Strategies

Overall, these results demonstrate that lower training loss or higher training PSNR does not necessarily correlate with better reconstruction quality. Instead, structured depth supervision particularly in the hybrid form acts as a powerful geometric prior that improves convergence stability, depth accuracy, and generalization to unseen views. The hybrid strategy emerges as the most effective and robust approach among all evaluated methods.

### 9.3 Qualitative Rendering Results

Qualitative evaluation is conducted by rendering novel views from unseen camera poses and visually comparing the outputs of all four strategies Baseline (NONE), Soft, Hard, and Hybrid against the ground-truth image. Figure 17 presents a side-by-side comparison for a representative validation frame, including corresponding error visualizations.

The **baseline NeRF (NONE)** produces visually plausible renderings at first glance; however, closer inspection reveals several structural deficiencies. The rendered object exhibits blurred boundaries, particularly along thin structures and edges, as well as semi-transparent regions and floating density artifacts around the base and articulated components. These issues are clearly reflected in the error visualization, where high-intensity error regions appear around object contours and depth discontinuities. This behaviour is consistent with the baseline’s reliance on purely photometric supervision, which provides insufficient geometric constraints.

The **soft depth-supervised strategy (SOFT)** shows noticeable improvements over the baseline. Object boundaries appear sharper, depth ordering between foreground and background components is more coherent, and floating artifacts are reduced. The corresponding error map exhibits lower overall intensity and smoother error transitions,

indicating improved geometric regularization. However, some residual blur and mild density spread remain, particularly in regions with complex geometry, reflecting the inherently weak enforcement of depth consistency in soft supervision.

The **hard-constrained strategy (HARD)** further improves geometric fidelity. Surfaces are more sharply localized, with clearer separation between the object and the supporting plane. Floating artifacts are largely eliminated, and fine structural details are better preserved. The error visualization confirms a reduction in high-error regions compared to both NONE and SOFT strategies. Nevertheless, slight over-sharpening and localized artifacts can be observed in some regions, likely caused by strict enforcement of geometric constraints in the presence of imperfect depth estimates.

The **hybrid strategy (HYBRID)** achieves the best overall qualitative performance. Rendered views closely match the ground truth in terms of surface sharpness, edge clarity, and global structure. The model successfully balances geometric precision with visual smoothness, avoiding both the ambiguity of the baseline and the rigidity of the hard-only approach. The error map exhibits the lowest intensity and most uniform distribution among all strategies, indicating superior reconstruction accuracy and visual consistency.

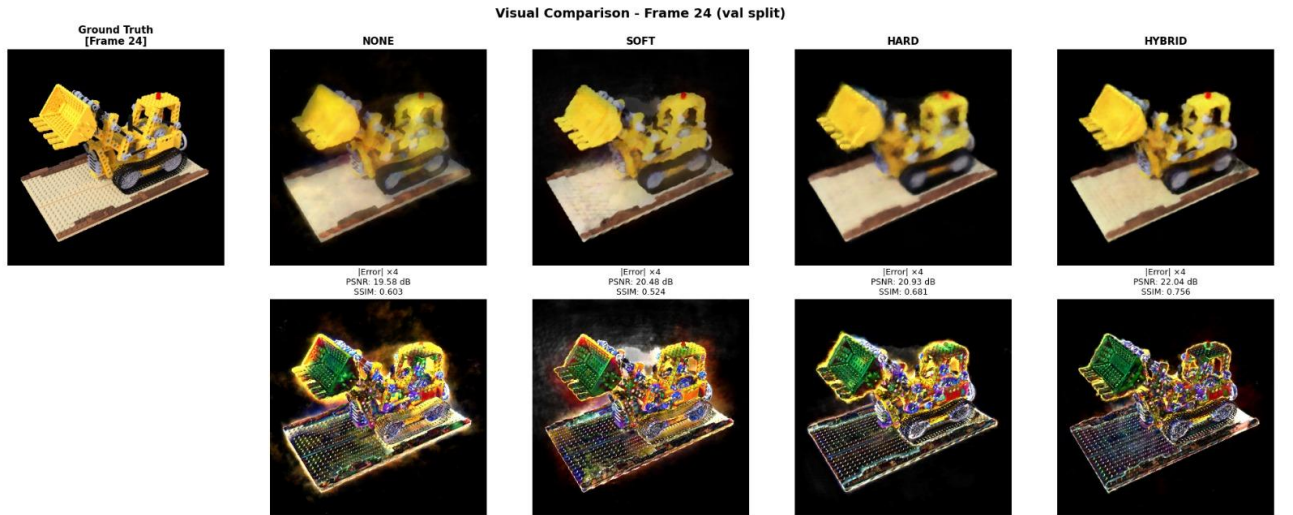


Figure 122 Qualitative visual comparison of novel view synthesis on the Lego dataset

Overall, the qualitative results strongly corroborate the quantitative findings. As the strength and structure of depth supervision increase from NONE to SOFT, HARD, and finally HYBRID, the rendered images demonstrate progressively improved geometric coherence, reduced artifacts, and higher perceptual quality. The hybrid strategy emerges as the most effective approach, delivering reconstructions that are both visually realistic and geometrically faithful on unseen views.



## 10 Discussion

The comparative analysis across baseline (photometric-only), soft depth-supervised, hard-constrained, and hybrid NeRF models provides clear insights into how varying degrees of geometric supervision influence neural radiance field reconstruction. These observations are supported by quantitative metrics including PSNR, RGB reconstruction loss, and depth accuracy as well as qualitative evaluation of synthesized views and predicted depth maps.

### 10.1 Geometric Consistency and Depth Accuracy

A primary distinction among the evaluated strategies lies in their ability to recover geometrically consistent 3D structure. The baseline NeRF, trained exclusively with photometric supervision, can generate visually plausible renderings but frequently fails to infer accurate geometry. This limitation manifests as blurred or thick surfaces, floating density artifacts, and inconsistent depth estimates, particularly in texture less regions or under sparse view coverage.

The soft depth supervised NeRF partially alleviates these issues by introducing depth as a regularization signal. By softly penalizing deviations between predicted and measured depth, the model exhibits improved surface localization and reduced density ambiguity compared to the baseline. However, because depth acts only as a weak constraint, density may still spread along rays, and surfaces often remain diffused rather than sharply defined.

In contrast, the hard-constrained NeRF enforces explicit geometric rules that govern density placement along camera rays. Free-space constraints suppress unphysical density accumulation in front of observed surfaces, while surface-concentration constraints force density to peak sharply at measured depths. This results in significantly improved geometric fidelity, with well-localized surfaces and reduced depth ambiguity.

Quantitatively, hard-constrained and hybrid models consistently achieve lower depth L2 loss, indicating superior alignment with ground-truth geometry. Depth-guided sampling further enhances accuracy by allocating more samples near true surface locations, reducing the likelihood of missing thin or complex structures.

### 10.2 Convergence Behavior and Training Stability

The training dynamics vary substantially across supervision strategies. The baseline NeRF exhibits slow and unstable convergence, as reflected by gradual and sometimes oscillatory PSNR improvement. This behavior arises from the ill-posed nature of photometric-only supervision, where multiple density configurations can yield similar RGB reconstructions, leading to ambiguous gradients during optimization.

The soft depth supervised model converges faster and more smoothly than the baseline due to the additional geometric signal. However, convergence remains slower than hard-constrained approaches, as the depth term does not strictly restrict density evolution.

The hard-constrained NeRF demonstrates faster convergence and smoother PSNR trajectories, as strong geometric priors reduce ambiguity and guide density formation early in training. Validation results indicate that these improvements generalize well to unseen views, confirming the effectiveness of explicit geometric supervision.

Nevertheless, purely hard constraints can introduce sensitivity to depth noise and instability during early optimization stages, particularly when initial density estimates are poor.

### 10.3 Trade-offs Between Flexibility and Geometric Rigidity

Each depth supervision strategy introduces a distinct trade-off between model flexibility and physical fidelity.

- **Baseline NeRF** retains maximum flexibility, allowing the network to explore multiple density configurations. While this can be advantageous when depth data is unavailable or unreliable, it often results in poor geometric accuracy.
- **Soft supervision** offers a compromise, improving stability while preserving adaptability to uncertain depth measurements.
- **Hard constraints** prioritize geometric correctness but may over-penalize errors when depth measurements are noisy or incomplete, potentially destabilizing early training.
- **Hybrid NeRF** balances these extremes by combining soft regularization with hard constraints, enabling robust learning while maintaining strong geometric enforcement.

This analysis highlights that no single strategy is universally optimal; rather, the choice depends on the reliability of depth data and the geometric precision required by the target application.

### 10.4 Hybrid Strategy and Sampling Design

The hybrid NeRF model integrates both soft depth supervision and hard geometric constraints, complemented by a hybrid sampling strategy in which approximately 50% of ray samples are depth-guided and 50% are uniformly distributed.

- **Depth-guided samples** exploit known surface locations, focusing computational effort where accurate density estimation is most critical.
- **Uniform samples** preserve exploration capacity along the ray, preventing overfitting to noisy or missing depth measurements.

Experimental results demonstrate that this hybrid approach avoids under-sampling artifacts, improves coverage of occluded or thin structures, and achieves more stable convergence than purely depth-guided or purely uniform sampling strategies.



## 10.5 Impact on Photorealistic Rendering Quality

Beyond geometric accuracy, depth supervision also influences visual fidelity. The baseline NeRF often exhibits semi-transparent artifacts, ghosting, and blurred edges, particularly in texture less regions. Soft supervision reduces these artifacts to some extent but does not fully eliminate them.

Both hard-constrained and hybrid NeRF models produce sharper surfaces and cleaner renderings by suppressing floating densities and enforcing physically meaningful structure. Importantly, the hybrid model maintains high photometric accuracy while avoiding excessive rigidity, demonstrating that physical geometric priorities can complement photometric supervision rather than conflict with it.

## 11 Research Contributions

This report makes the following key contributions to the study of depth supervision strategies in Neural Radiance Fields:

1. A systematic analysis of baseline NeRF with purely photometric supervision, identifying its strengths in visual synthesis and its inherent limitations in resolving geometric ambiguity and surface localization.
2. A clear conceptual and mathematical exposition of soft depth supervision, demonstrating how probabilistic depth guidance improves training stability while retaining flexibility under noisy or sparse depth conditions.
3. A rigorous formulation and interpretation of hard geometric constraints, including free-space enforcement and surface concentration, within the volumetric rendering and density optimization framework.
4. A structured comparative evaluation of baseline, soft-constrained, and hard-constrained NeRF variants, highlighting their effects on optimization behaviour, density sharpness, convergence speed, and geometric accuracy.
5. An in-depth discussion of the trade-offs between flexibility and rigidity in NeRF training, particularly under imperfect depth supervision and real-world sensing constraints.
6. A unified analytical narrative that connects physical intuition, mathematical modelling, and observed reconstruction behaviour, serving as a reference foundation for future research on hybrid depth supervision strategies in NeRF-based 3D reconstruction.

## 12 Conclusion

In this study, we conducted a comprehensive comparison of baseline NeRF, soft depth-supervised NeRF, and hard-constrained NeRF to examine how different supervision strategies influence volumetric scene reconstruction. Baseline NeRF, driven solely by photometric consistency, can generate visually plausible novel views under dense multi-view conditions; however, it often exhibits geometric ambiguities such as floating densities, blurred surfaces, and depth inconsistencies, particularly in texture less or sparsely observed regions.

Incorporating soft depth supervision introduces probabilistic geometric guidance that improves depth consistency and training stability while maintaining robustness to noise and incomplete measurements. This strategy strikes a balance between visual fidelity and geometric regularization but does not fully eliminate volumetric ambiguity. In contrast, hard-constrained NeRF explicitly enforces physically motivated geometric rules, such as free-space constraints and surface density concentration, leading to sharper surfaces, improved geometric fidelity, and faster convergence. These benefits come at the cost of increased sensitivity to depth noise, as inaccurate measurements can overly restrict the optimization process.

The comparative analysis reveals a fundamental trade-off among the three strategies: baseline NeRF offers maximum flexibility with minimal supervision, soft constraints provide robustness and moderate geometric improvement, and hard constraints deliver precise surface reconstruction when reliable depth information is available. Overall, the results highlight the importance of integrating photometric and geometric supervision to achieve robust, physically consistent, and high-quality scene representations. This work establishes a solid foundation for future exploration of hybrid supervision frameworks that adaptively combine soft and hard constraints for scalable and real-world NeRF applications in robotics, augmented reality, and digital twin systems.

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